

Advanced (Carolina Reaper)

Q1. Solve for x in the equation $2x + 5 = 11$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. A book costs

15

more than a pencil. If x represents the cost of the pencil, the equation is $x + 15 = 25$. Solve for x

- (A) $x = 10$
- (B) $x = 12$
- (C) $x = 15$
- (D) $x = 18$
- (E) $x = 20$

Q3. Solve the equation $4x - 2 = 2x + 6$

- (A) $x = 2$
- (B) $x = 4$
- (C) $x = 6$
- (D) $x = 8$
- (E) $x = 10$

Q4. Tom has been saving money for a new bike and has x dollars. He wants to buy a bike that costs 120. If he has already saved 40, the equation is $x + 40 = 120$. Solve for x

- (A) $x = 60$
- (B) $x = 70$
- (C) $x = 80$
- (D) $x = 90$
- (E) $x = 100$

Q5. Solve for y in the equation $y - 3 = 7$

- (A) $y = 4$
- (B) $y = 5$
- (C) $y = 10$
- (D) $y = 11$
- (E) $y = 12$

Q6. A group of friends want to split a bill evenly. If the total bill is

48

and there are x friends, the equation is $48 = 8x$. Solve for x

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = 6$
- (D) $x = 7$
- (E) $x = 8$

Q7. Solve the equation $2x + 2 = 5x - 3$

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q8. Ashley has been practicing a new song on her guitar. If she practices for x hours per day and wants to practice for a total of 12 hours, the equation is $3x = 12$. Solve for x

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Open Ended Questions (FRQ)

Q9. Solve the equation $x/4 + 2 = 9$ and check your solution for accuracy. Show all work

Q10. Solve the equation $2x - 5 = 11$ and then use the solution to find the value of $3x + 2$. Show all work

Chef's Tips

- Read each question carefully
- Use a pencil to write your answers
- Check your work